Subject: Programming with Python **Subject Code:** 01CT1309

Date: 06/09/2025 **Timing: 9:30 am to 4:15 pm**

Long Hour Coding Exam

Task: Design and Implementation of a Python Package signal_ICT_StudentName_EnrollmentNo for Signal Generation and Operations.

Problem Statement:

You are required to design a custom Python package named signal_ICT_StudentName_EnrollmentNo that demonstrates fundamental concepts of Signals and Systems. The package must be modular, containing three separate modules:

1. **unitary_signals.py**

   Implement the following functions:
   - unit_step(n) – Generates a unit step signal.
   - unit_impulse(n) – Generates a unit impulse signal.
   - ramp_signal(n) – Generates a ramp signal.

   Each function should return a NumPy array and plot the signal using matplotlib.
2. **trigonometric_signals.py**

   Implement the following functions:
   - sine_wave(A, f, phi, t) – Generates a sine wave with amplitude A, frequency f, phase phi, and time vector t.
   - cosine_wave(A, f, phi, t) – Generates a cosine wave with similar parameters.
   - exponential_signal(A, a, t) – Generates an exponential signal.
3. **operations.py**

   Implement the following signal operations:
   - time_shift(signal, k) – Shifts the signal by k units.
   - time_scale(signal, k) – Scales the time axis of the signal by factor k.
   - signal_addition(signal1, signal2) – Performs addition of two signals.
   - signal_multiplication(signal1, signal2) – Performs point-wise multiplication of two signals.

Main Script (main.py)

- Import the above modules from the package.
- Demonstrate the following tasks:
  1. Generate and plot a unit step signal and a unit impulse signal of length 20.
  2. Generate a sine wave of amplitude 2, frequency 5 Hz, phase 0, over t = 0 to 1 sec.
  3. Perform time shifting on the sine wave by +5 units and plot both original and shifted signals.
  4. Perform addition of the unit step and ramp signal and plot the result.
  5. Multiply a sine and cosine wave of same frequency and plot the result.

Expected Deliverables:

1. Folder structure of the package:
2. signal_ICT_StudentName_EnrollmentNo /
3. __init__.py
4. unitary_signals.py
5. trigonometric_signals.py
6. operations.py
7. main.py
8. Well-documented Python code with function definitions and comments.
9. Proper use of NumPy (for signal arrays) and Matplotlib (for plotting).
10. At least 3 plots showing signals and operations as per requirements.

Student Submission Requirements:

- The Wheel file (.whl) and source distribution (.tar.gz) inside a dist/ folder.
- A README.md explaining package modules, installation, and usage.
- A screenshot/PDF report showing:
    1. Successful local installation from wheel.
    2. Successful upload to TestPyPI.
    3. Successful installation from TestPyPI.
- GitHub repo link. https://github.com/Harshalbhatt2806/Python-LHC

Reference Link: https://www.youtube.com/watch?v=9Ii34WheBOA

Evaluation Criteria rubric (20 Marks):

- Package Design & Modularity (5 Marks) – Proper structure with three modules.
- Correct Implementation of Signals (5 Marks) – Unitary & trigonometric signals.
- Correct Implementation of Operations (5 Marks) – Scaling, shifting, addition, multiplication.
- Main Script Demonstration (3 Marks) – Calling and plotting from all modules.
- Code Quality & Documentation (2 Marks) – Comments, readability, efficiency.

Subject: Programming with Python **Subject Code:** 01CT1309

Student Name:HARSHAL BHATT **Enrollment No:92510133020**

Division: 3EK3

```
PS E:\MU\PWP\signal_ICT_HARSHAL_92510133020> python -m build
* Creating isolated environment: venv+pip...
* Installing packages in isolated environment:
  - hatchling
* Getting build dependencies for sdist...
* Building sdist...
* Building wheel from sdist
* Creating isolated environment: venv+pip...
* Installing packages in isolated environment:
  - hatchling
* Getting build dependencies for wheel...
* Building wheel...
Successfully built signal_ict_harshal_92510133020-0.0.1.tar.gz and signal_ict_harshal_92510133020-0.0.1-py3-none-any.whl
```

```
Requirement already satisfied: certifi>=2017.4.17 in c:\users\harshal bhatt\appdata\local\programs\python\python313\lib\site-packages (from requests>=2.20->twine) (2025.7.14)
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\users\harshal bhatt\appdata\local\programs\python\python313\lib\site-packages (from rich>=12.0.0->twine) (4.0.0)
Requirement already satisfied: mdurl~=0.1 in c:\users\harshal bhatt\appdata\local\programs\python\python313\lib\site-packages (from markdown-it-py>=2.2.0->rich>=12.0.0->twine) (0
.1.2)
Requirement already satisfied: more-itertools in c:\users\harshal bhatt\appdata\local\programs\python\python313\lib\site-packages (from jaraco.classes->keyring>=21.2.0->twine) (1
0.8.0)
PS E:\MU\PWP\signal_ICT_HARSHAL_92510133020> python -m twine upload dist/*
Uploading distributions to https://upload.pypi.org/legacy/
WARNING  This environment is not supported for trusted publishing
Enter your API token:
Uploading signal_ict_harshal_92510133020-0.0.1-py3-none-any.whl
100% ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 359.2/359.2 kB • 00:01 • 285.3 kB/s
Uploading signal_ict_harshal_92510133020-0.0.1.tar.gz
100% ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 352.6/352.6 kB • 00:00 • 31.5 MB/s

View at:
https://pypi.org/project/signal-ICT-HARSHAL-92510133020/0.0.1/
PS E:\MU\PWP\signal_ICT_HARSHAL_92510133020>
```

```
PS C:\Users\ASUS\Desktop\Shivang> pip install signal_ICT_HARSHAL_92510133020
Defaulting to user installation because normal site-packages is not writeable
Collecting signal_ICT_HARSHAL_92510133020
  Downloading signal_ict_harshal_92510133020-0.0.1-py3-none-any.whl.metadata (505 bytes)
Downloading signal_ict_harshal_92510133020-0.0.1-py3-none-any.whl (356 kB)
Installing collected packages: signal_ICT_HARSHAL_92510133020
Successfully installed signal_ICT_HARSHAL_92510133020-0.0.1
```